

IBM University Engagement Project Submission

**“Exploring the Power of Agentic AI with
IBM Granite and LangFlow”**

Project Title –

LearnMate AI - Personalized Course Pathways

Domain of Project – Education / EdTech

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Problem Statement: LearnMate AI

The Challenge

Students often struggle to identify the right learning path that aligns with their interests and long-term goals due to the overwhelming number of online courses and a lack of personalized guidance.

The Objective

Build an AI-powered coach that interacts with students, understands their interests (like Frontend Development, Cybersecurity, etc.), assesses their current skill level, and dynamically builds a personalized course roadmap that adapts over time.

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used: LangFlow + IBM watsonx.ai + Granite Models

Additional Technology Used:

Frontend: React.js, Tailwind CSS

Backend: Node.js, Express.js (for secure API routing)

Proposed Solution: AI-Powered Career Coach

The proposed solution is a secure, AI-powered career coach built using Langflow, IBM watsonx.ai, and Granite Models to provide instant, personalized course pathways.

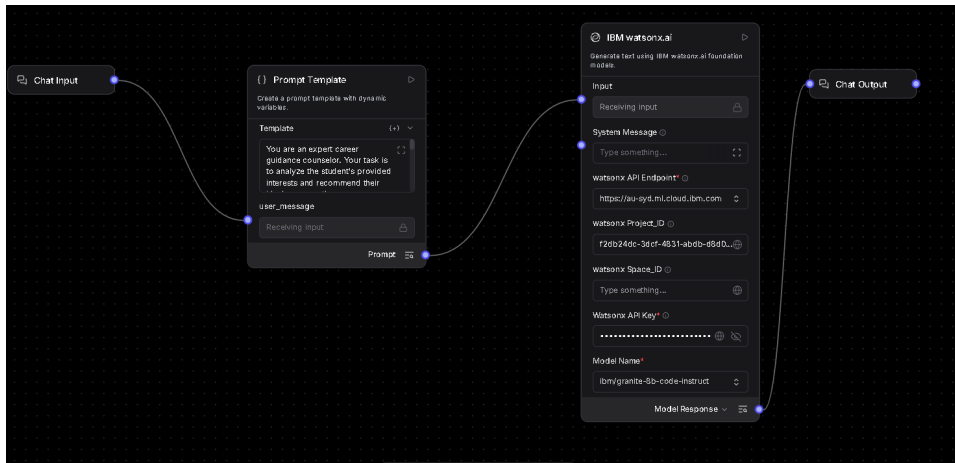
At its core, the solution captures user inputs (hobbies, goals) via a React interface. This data is securely routed through a Node.js backend to protect API credentials, ensuring enterprise-grade security.

The system processes this input through a strict Prompt Template in Langflow, utilizing the `ibm/granite-8b-code-instruct` model. The AI acts as an expert counselor, reasoning through the student's unique profile to generate a structured 3-part diagnostic report (Top Recommendations, Reasoning, and Next Steps).

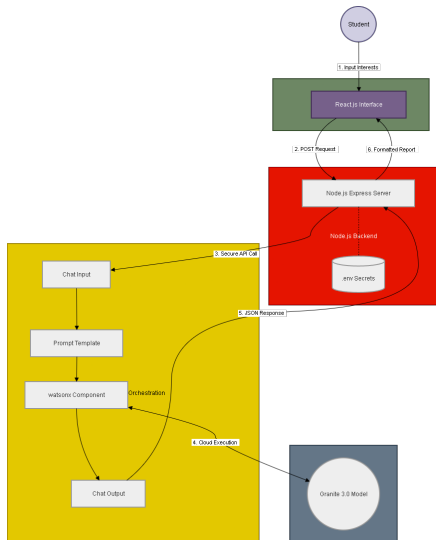
Langflow Components Used

1. **Chat Input** – Interface where the React frontend sends the student's message regarding hobbies, subjects, and goals.
2. **Prompt Template** – Structures user input into a strict request for a diagnostic roadmap, ensuring the AI stays on topic.
3. **IBM watsonx AI Component** – Connects the Langflow logic to the secure cloud environment for model execution.
4. **IBM Granite Model** – `ibm/granite-8b-code-instruct` → Fast, efficient model chosen for strict formatting and real-time generation.
5. **Chat Output** – Captures the final structured text generated by the AI and routes it back to the Node.js backend to display to the user.

Langflow Workflow



Architecture Blueprint



Role of Agentic AI in LearnMate

Agentic AI enables LearnMate to function as an intelligent, context-aware coach rather than a simple Q&A chatbot.

Using Langflow and IBM Granite models, the system autonomously parses complex, unstructured student profiles into actionable roadmaps.

It brings context awareness by differentiating between domain specifics (e.g., Frontend vs. Backend) to align with distinct user goals. The agent dynamically adapts its response based on the student's perceived current skill level, acting proactively rather than reactively.

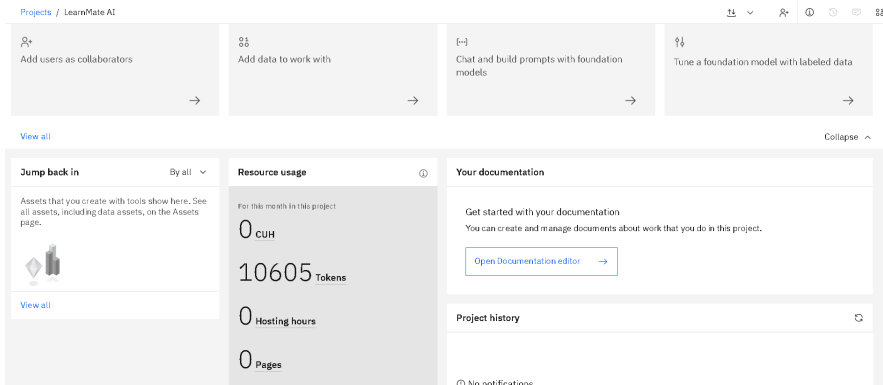
Technology Used

- **Langflow platform** – For designing and orchestrating the user-to-AI data flow visually.
- **IBM Granite model** – e.g., `ibm-granite-8b-code-instruct` – For natural language understanding, strict prompt adherence, and personalization.
- **IBM Cloud** – Provides the infrastructure to build, deploy, and scale the application efficiently.
- **Node.js / Express** – Acts as a secure middleware bridge to protect API credentials.
- **IBM Watsonx.ai** – For secure model deployment and AI governance.

IBM Cloud Token Usage

Resource Monitoring in watsonx.ai

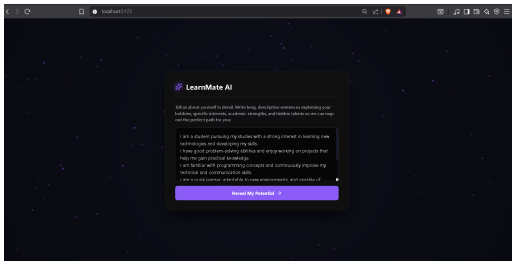
Project token usage is actively monitored. This metric tracks the number of input tokens (prompts) and output tokens (generated text) processed securely by the Granite model.



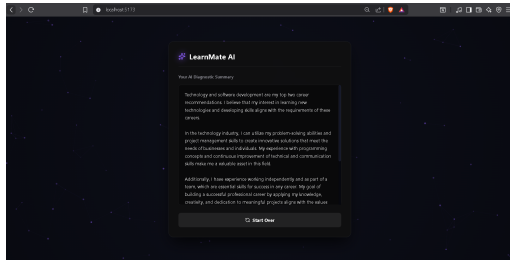
Results

System Output & User Dashboard

User Input / Dashboard



AI Generated Roadmap



Novelty and Uniqueness

- **Hyper-Personalization** – Tailors the output specifically to the student's unique combination of hobbies, current skill level, and goals rather than generic searches.
- **Secure Architecture** – Uses a Node.js middleware bridge to protect IBM Cloud credentials, ensuring they are never exposed to the client.
- **Actionable Diagnostics** – Doesn't just give answers; provides a strict 3-part report with actual actionable "Next Steps".
- **Scalable & Enterprise-Ready** – Built on IBM watsonx.ai, allowing for future expansion into fetching live course data.

GitHub Repository

Please create public github repository (Enable Add README button while creating repository)

GitHub URL –

`https://github.com/aibeljoy199/learnmate-ai`

Make sure you have Uploaded following files into your github repository:

1. `app.json` file
2. `yourproblemstatement.pdf` file
3. `your projectpresentation.pdf` file
4. Full codebase (Frontend & Backend folders)

Future Scope

1. Live Course API Integration

Connecting the AI to real-time educational platforms (like Coursera or edX) to fetch and recommend exact course URLs and prices dynamically.

2. Progress Tracking & Authentication

Adding a secure database to track the student's progress along their personalized learning path over time and adapt recommendations as they level up.

3. Multi-Modal Resume Analysis

Allowing students to upload their resumes (PDFs) so the AI can automatically parse their current skill levels without requiring manual text input.

Thank You!

Thank you for your time and interest.